Homework_3_Report

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1 Introduction

The goal of this report is to evaluate the use of singular value decomposition for compressing grayscale images. This approach involves preprocessing an image, applying a block-wise SVD compression to the image, and produce important metrics and visualizations. By doing this we can ultimately determine the efficiency in terms of image quality of singular value decomposition.

2 Methodology

The first step in this process is to preprocess the image. I selected china.jpg from the scikit-learn dataset because its details make it easy to observe differences across rank-k approximations. The original image is a 427×640 RGB photograph, so I resized it to 256×256 to ensure both dimensions are divisible by 8, converted it to grayscale, and partitioned it into non-overlapping 8×8 blocks.

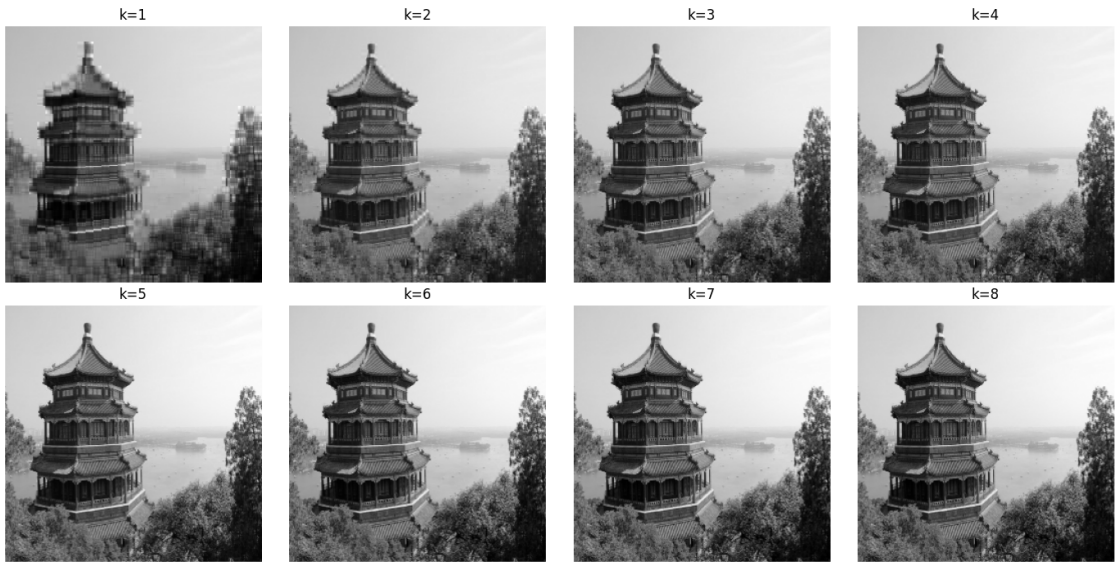
The second step is to implement a function that applies SVD to each 8×8 block, producing the matrices needed for compression and reconstruction. The third step extends this process across the entire image by applying SVD to every block and computing the evaluation metrics, specifically the compression ratio and the Frobenius norm of the reconstruction error for each value of k. The final

step is to analyze these metrics by visualizing them in plots, allowing us to observe how compression performance and reconstruction quality change as k varies.

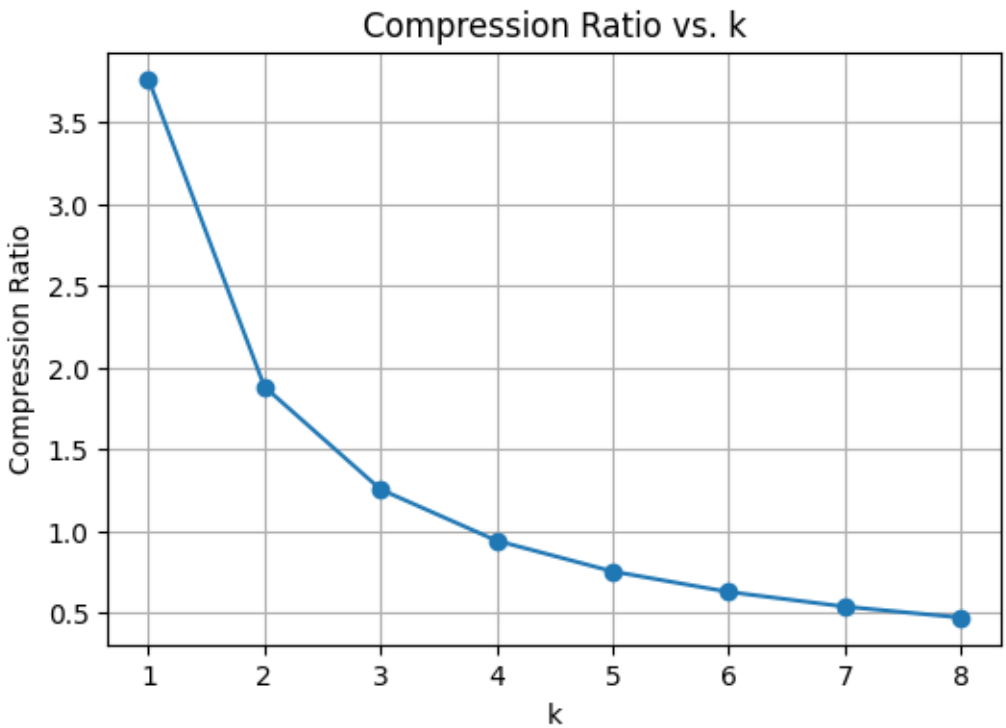
3 Results

In this report I came up with some key findings.

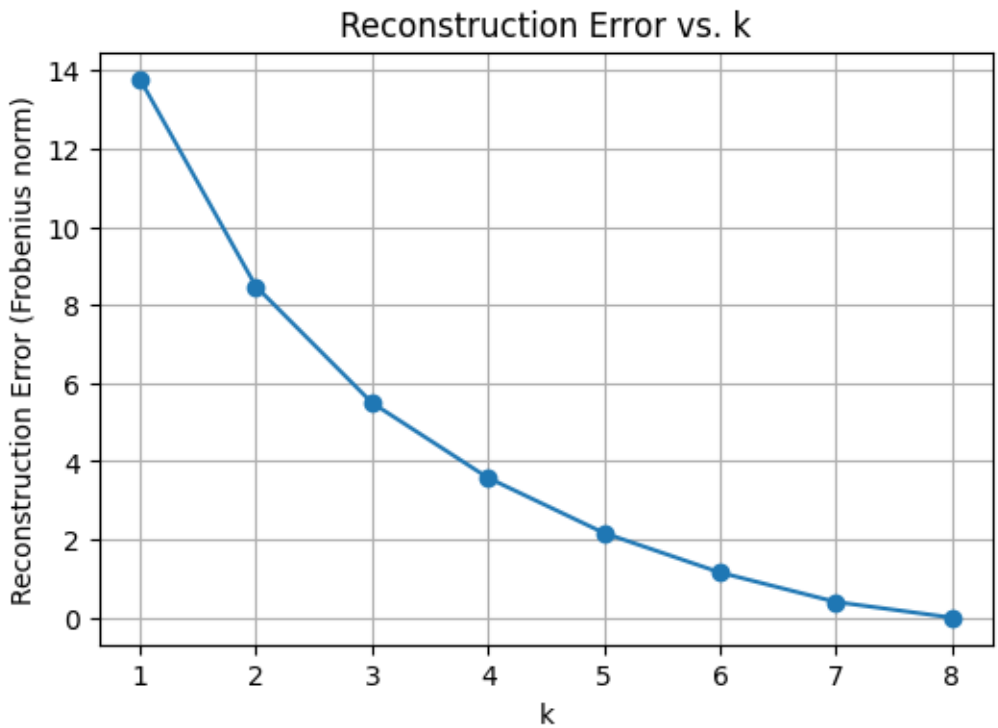
The image below shows the compressed and reconstructed output for each value of k. We can see a clear pattern. The reconstruction quality improves as k increases. Each additional singular value preserves more of the block's underlying structure, variation, and detail. Since the singular values encode the most important information in descending order of significance. By retaining more of these components, the reconstruction captures a larger portion of the block's true energy, leading to visibly sharper and more accurate results at higher k values.



The graph below shows the compression ratio as a function of k. It illustrates how the amount of data retained per block increases with each additional singular value. Since SVD stores 17*k values for an 8×8 block, larger values of k require storing more information, which reduces the overall compression ratio. As a result, the image becomes less compressed as k increases. This reflects the trade-off in SVD-based compression, which keeps more singular values to improve reconstruction quality but requires storing more data, while smaller k values achieve stronger compression at the cost of higher reconstruction error.



The plot below shows how the reconstruction error changes with k. It captures the pattern that emerges during block-wise reconstruction; the error decreases steadily as k increases. A larger k retains more singular values, which means more of the block's structure and information are preserved. Because SVD guarantees the best possible rank k approximation in the Frobenius norm, each additional singular value reduces the loss of information and improves the quality of the reconstruction. As a result, higher k values consistently produce lower error and more accurate approximations.



4 Conclusion

Completing this research helped me understand the practical value of SVD in image compression and dimensionality reduction. While SVD can become computationally expensive for large images, its ability to retain only the most informative components makes it a powerful tool for portability and efficient storage. For smaller images, the method performed especially well, as k increased, reconstruction error decreased, while compression ratios decreased. This behavior illustrates the core idea behind dimensionality reduction of preserving the top k components that capture the greatest variance in the data. Understanding this connection also explained why SVD is foundational in many visual learning models, where reducing complexity without losing essential structure is critical for performance.